

# KAZI SAMIN NAWAL

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## PROFESSIONAL SUMMARY

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I am a MERN stack developer with strong web, database, and cloud skills, focused on building scalable, user-friendly applications. I also design and develop AI tools, chatbots, and AI agents, applying intelligent systems to solve real-world problems through code.

## EDUCATION

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B.Sc. in Computer Science & Engineering

2022-2026

Independent University, Bangladesh

CGPA: 3.85

## TECHNICAL SKILLS

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- **Frontend:** HTML5, CSS3, TypeScript, JavaScript, React.js, Tailwind CSS, Next.js
- **Backend:** Node.js, Express.js, Flask, FastAPI, JWT
- **Database:** MongoDB, MySQL, PostgreSQL
- **Machine Learning Frameworks:** Tensorflow, Pytorch, Scikit-learn, Pandas, NumPy
- **Tools & Technologies:** Python, Firebase, WebSockets, Git, Java
- **LLM & AI Application Tools :** LangChain, Vector Databases , RAG Pipelines, Prompt Engineering

## EXPERIENCE

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Data & Design Lab, University of Dhaka

August 2026 - Present

*Full Stack Software Developer*

- Developed a Bangla AI chatbot for accessing electricity-related information through natural voice and text, achieving an SUS score of 84.9.
- Developed the chatbot backend with Python and Flask, connecting LLMs with speech-to-text (STT) and text-to-speech (TTS) for voice-based conversations.

*Research Assistant*

October 2025 - July 2026

- Developed and evaluated machine learning models for healthcare insurance fraud detection using Python and Jupyter Notebook, analyzing 50.5 million real-world records.
- Worked with researchers and developers in an Agile team, using Git for version control and participating in code reviews.

Independent University, Bangladesh

January 2025 - October 2025

*Teaching Assistant*

- Conducted classes and hands-on workshops on HTML, CSS, JavaScript, modern web frameworks and AI technologies.
- Assisted students with practical implementation, debugging, and project development.

## PROJECTS

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1. **DnD Lab Workspace** | Next.js, FastAPI, PostgreSQL, Docker | [Live Application](#) | [Github](#)
  - Built a social platform for research teams to share updates, manage tasks, and collaborate on files in a centralized workspace.
  - Developed task tracking, file sharing, and team collaboration features to streamline research coordination.
  - Used TypeScript and Python for frontend development, backend services, and automation.
2. **Platemate**  | React, Node.js, Express, MongoDB, Firebase | [Github](#)
  - Secure seller onboarding and product management.
  - Real-time cart updates and order placement.
  - Interactive UI with meal categories and diet plans.
3. **Pantrypal**  | Next.js, Node.js, Express, MongoDB, Groq AI | [Client Github](#) | [Server Github](#)
  - Automated pantry tracking with real-time freshness insights and expiry alerts.
  - AI-driven chatbot for natural language shopping, cart management, and recipe suggestions.
  - Flexible subscription bundle builder with secure order processing and dynamic inventory updates.
4. **MediCloud**  | React, Node.js, Express, MongoDB | [Client Github](#) | [Server Github](#)
  - Multi-Role Authentication System. Firebase authentication for secure login and registration.
  - Dynamic Product Management
  - Admin, seller and user dashboard

## PUBLICATIONS

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1. H. Mekat, J. Hossain, **K. Nawal**, M. Alam, A. Ema, S. Ahmed, M. Hasan, “Diabetes Management with Automated Health Monitoring and Diet Recommendations for Patients,” IEEE SERA 2025, pp. 309–314, 2025. DOI: 10.1109/SERA65747.2025.11154527
2. M. J. H. Mekat, **K. S. Nawal**, et al., “AI-Driven Persona Generation for User-Centered Design Research: Evaluating Quality and Consistency Across Large Language Models,” ICMCSI, 2026, pp. 1918–1925. DOI: 10.1109/ICMCSI67283.2026.11412817
3. M. J. H. Mekat, **K. S. Nawal**, et al., “Deep Learning-Based Stenosis Segmentation in X-ray Angiography: Vision Transformers vs. CNNs,” ICCIT, 2025, pp. 3553–3558. DOI: 10.1109/ICCIT68739.2025.11490502

## CERTIFICATIONS

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- Complete Web Development Course- Programming Hero ( March 2025)
- CCNA: Introduction to Networks (April 2024)